Sleep Disturbances, Pain, and Depressive Symptoms as Predictors of 4-Year Cognitive Scores in Older U.S. Adults: An XGBoost Analysis

Utpol Das, MPH (1); José Eduardo Cabrero-Castro, PhD (2); Sadaf Arefi Milani, PhD, MPH (3)
(1) Division of Palliative Medicine, Department of Internal Medicine, The University of Texas Medical Branch at Galveston, Galveston, Texas
(2) Department of Public Health Sciences, College of Health Sciences, The University of Texas at El Paso, El Paso, Texas
(3) Department of Epidemiology, The University of Texas Medical Branch at Galveston, Galveston, Texas
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Key finding: frequent trouble falling asleep, waking too early, and moderate-to-severe pain marked lower predicted cognitive scores 4 years later, but sleep, pain, and depressive symptoms together accounted for only 6% of the model's predictive weight. Past cognition, age, and education carried 85% of the predictive weight.

In plain words: we followed 10,166 adults aged 50 and older in the Health and Retirement Study for 4 years. People who had trouble falling asleep or woke too early most of the time, or who had moderate or severe pain, had slightly lower predicted memory and thinking scores 4 years later. But these symptoms, together with depressive symptoms, accounted for only a small share (about 6%) of what drove the model's predictions. A person's earlier score, age, and education mattered far more.

Background

- Depressive symptoms, sleep problems, and pain are common in later life, often occur together, and can be treated.
- Each has been linked to later cognitive decline, but most studies look at one symptom at a time, often as a summary score.
- It is unclear how much these symptoms add to predicting later cognition once past cognition, age, and education are known, and at what symptom level any contribution appears.

Objectives

1. Quantify how much depressive symptoms, sleep disturbances, and pain contribute to predicting cognitive scores in 2022.
2. Identify which sleep and pain items contribute most.
3. Describe the symptom levels at which these contributions arise.

Methods

- **Data:** Health and Retirement Study, a national panel of U.S. adults older than 50. Predictors were measured in 2018 and the outcome in 2022 (4-year horizon). 10,166 participants had cognitive scores in both waves and complete 2018 data.
- **Outcome:** Imputed 27-point cognitive score in 2022 (immediate and delayed word recall, serial 7 subtraction, and backward counting). Higher scores mean better performance.

- **Symptoms (2018):** 8-item CES-D depressive symptom score (0 to 8); four sleep items (trouble falling asleep, waking during the night, waking too early, feeling rested), each rarely or never, sometimes, or most of the time; three pain items (frequent pain, severity, interference with usual activities).
- **Other predictors:** 2018 cognitive score, age, years of education, sex, race and ethnicity, marital status, and rural residence.
- **Model:** Extreme gradient boosting (XGBoost) regression, tuned with fivefold cross-validation in a training sample (n = 5,084) and evaluated in a held-out test sample (n = 5,082).
- **Explanation:** Shapley additive explanation (SHAP) values describe how many points each predictor moved a person's predicted 2022 score compared with the average prediction. Sleep and pain items were also summed into domains. SHAP values describe the model, not cause and effect.

Results

- Mean age was 66.2 years; 60.1% were women, 23.2% non-Hispanic Black, and 16.8% Hispanic.
- Baseline cognition, age, and education accounted for 85.0% of total SHAP importance. Sleep disturbances (3.38%), pain (1.75%), and depressive symptoms (0.95%) together accounted for 6.1%.
- Within symptoms, trouble falling asleep (1.93%), pain severity (1.45%), and waking too early (1.30%) ranked highest; waking during the night (0.34%) contributed little.
- Contributions were near zero or slightly positive at low symptom levels and negative only at the highest levels.

Model performance in the test sample: R-squared 0.447 (44.7% of variance explained); root mean squared error 3.42 points; mean absolute error 2.66 points.

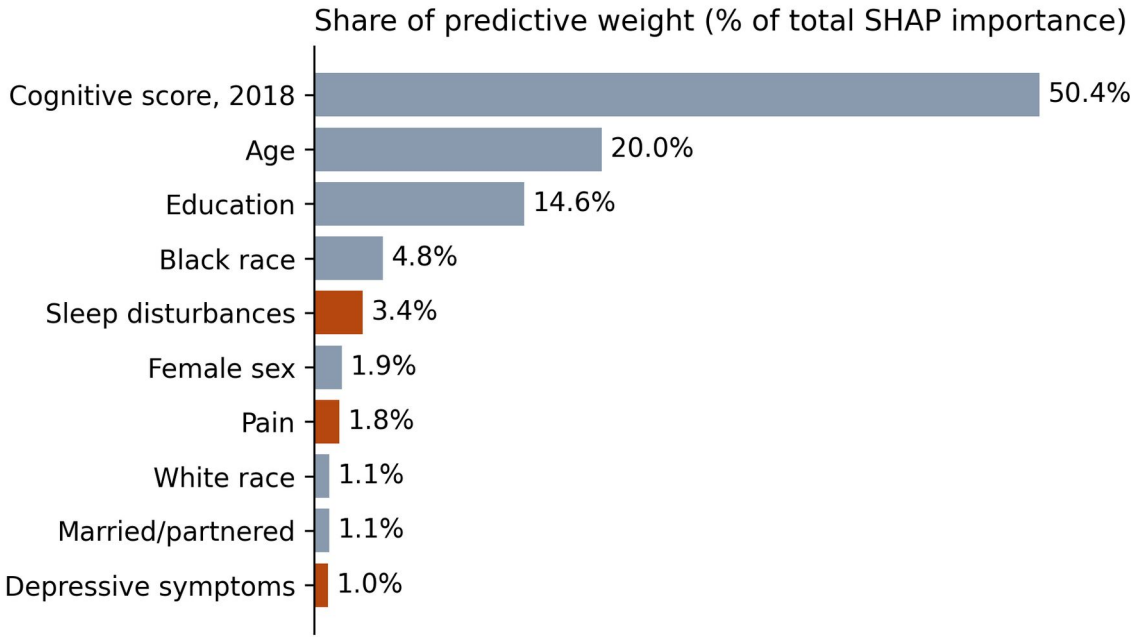


Figure 1. Horizontal bar chart of each predictor's share of total SHAP importance: 2018 cognitive score 50.4%, age 20.0%, education 14.6%, Black race 4.8%, sleep disturbances 3.4%, female sex 1.9%, pain 1.8%, White race 1.1%, married or partnered 1.1%, depressive symptoms 1.0%. The three symptom domains are highlighted.

Table 1. Share of predictive weight for the 2022 cognitive score (top 10 predictors, test sample)

Rank	Predictor or domain	Mean absolute SHAP, points (95% CI)	Share of total importance
1	Cognitive score, 2018	1.93 (1.89 to 1.97)	50.4%
2	Age	0.765 (0.748 to 0.783)	20.0%
3	Years of education	0.559 (0.547 to 0.570)	14.6%
4	Black race (non-Hispanic)	0.183 (0.179 to 0.186)	4.77%
5	Sleep disturbances (domain)	0.130 (0.126 to 0.133)	3.38%
6	Female sex	0.0741 (0.0730 to 0.0753)	1.93%
7	Pain (domain)	0.0669 (0.0655 to 0.0684)	1.75%
8	White race (non-Hispanic)	0.0405 (0.0401 to 0.0409)	1.06%
9	Married or partnered	0.0403 (0.0395 to 0.0412)	1.05%
10	Depressive symptoms	0.0364 (0.0358 to 0.0370)	0.95%

Table 2. Mean SHAP value by symptom level (points on the 27-point score; negative values lower the predicted score)

Symptom	Lower levels	Highest level(s)
Trouble falling asleep	Rarely 0.041; sometimes 0.042	Most of the time −0.254
Waking too early	Rarely 0.044; sometimes −0.002	Most of the time −0.175
Waking during the night	Rarely 0.009; sometimes 0.004	Most of the time −0.020
Feeling rested on waking	Most of the time 0.010; sometimes 0.020	Rarely or never −0.097
Pain severity	None 0.041; mild 0.045	Moderate −0.064; severe −0.163
Pain interference	None 0.009; low 0.012	High −0.037
Depressive symptoms (CES-D)	Score 0: 0.024; score 1: 0.031	Score 2: −0.039; scores 3 to 8: −0.057 to −0.062

What it means

- Four-year cognitive predictions depended mainly on baseline cognition, age, and education.
- Frequent difficulty falling asleep or waking too early, moderate or severe pain, and even a few depressive symptoms were associated with lower predicted scores.
- The largest symptom contribution (0.25 points, about 0.055 SD) was far smaller than the average prediction error (3.42 points), so symptoms alone should not be expected to identify future decline.
- These symptoms warrant clinical attention in their own right.

Limitations

- This is a prediction study; SHAP values describe the fitted model and are not causal effects.
- One random training and test split; no baseline-only comparison, calibration, or external validation.
- Complete-case sample of people who took part in both waves; survey weights were not used, so results may not represent all U.S. adults aged 50 and older.
- Symptoms were self-reported once; the CES-D includes a restless-sleep item that overlaps with the sleep items.

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Acknowledgments and contact

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Contact: Utpol Das, MPH, utdas@utmb.edu. ORCID 0009-0001-7672-4473.